Survey

The Quest for the Right Mediator: Surveying Mechanistic Interpretability Through the Lens of Causal Mediation Analysis

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Interpretability provides a toolset for understanding how and why neural networks behave in certain ways. However, there is little unity in the field: most studies employ ad-hoc evaluations and do not share theoretical foundations, making it difficult to measure progress and compare the pros and cons of different techniques. Furthermore, while mechanistic understanding is frequently discussed, the basic causal units underlying these mechanisms are often not explicitly defined. In this article, we propose a perspective on interpretability research grounded in causal mediation analysis. Specifically, we describe the history and current state of interpretability taxonomized according to the types of causal units (mediators) employed, as well as methods used to search over mediators. We discuss the pros and cons of each mediator, providing insights as to when particular kinds of mediators and search methods are most appropriate. We argue that this framing yields a more cohesive narrative of the field and helps researchers select appropriate methods based on their research objective. Our analysis yields actionable recommendations for future work, including the discovery of new mediators and the development of standardized evaluations tailored to these goals.

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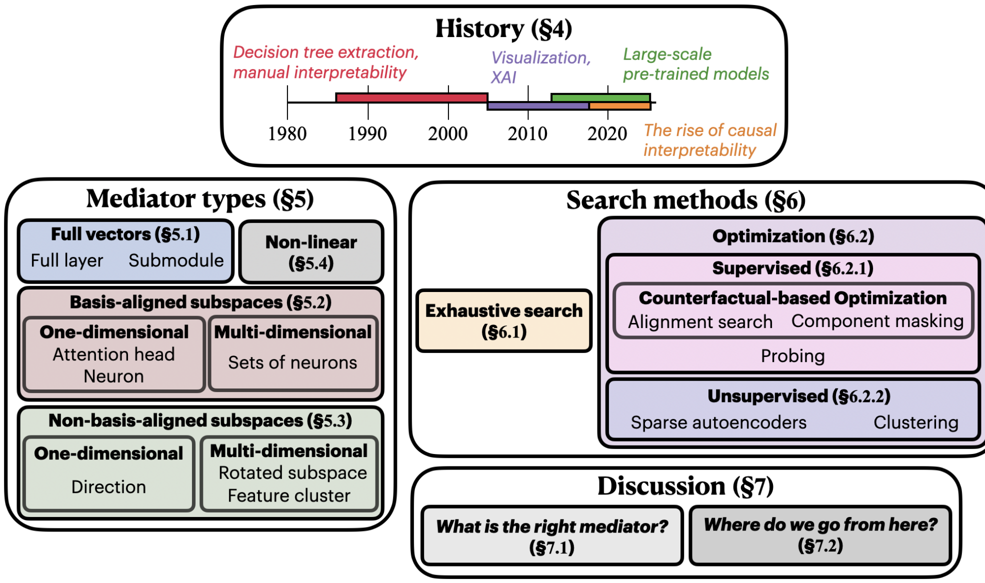

Figure 1
Outline of survey. We first define necessary causal terminology (§2) and contextualize our perspective with others' (§3). We then give an overview of the history of mechanistic interpretability centered on units of causal analysis (§4). We then survey and categorize commonly used units of analysis and describe their strengths and weaknesses (§5), as well as methods for searching over them (§6). Finally, we discuss (§7) what we consider to be among the most important questions in mechanistic interpretability: **What are the right causal abstractions for understanding and discussing the inner workings of NNs** (§7.1)? **What kinds of mediators and research will be needed to advance the field** (§7.2)?

1. Introduction

To understand how a neural network (NN) will generalize, we must understand the causes of its behavior. These causes include inputs, but also the intermediate computations of the network; this survey is concerned with understanding these intermediate computations. How can we understand what a NN's computations represent, such that we can obtain a deeper algorithmic understanding of *how* and *why* models behave the way they do? For example, if a language model (LM) decides to refuse a user's request, was the refusal mediated by an underlying concept of toxicity, by the presence of superficial correlates of toxicity (such as the mention of particular demographic groups), or some other unexpected variable? The former would be significantly more likely to robustly and safely generalize. These questions motivate the field of mechanistic interpretability (MI), which aims to understand how NNs arrive at particular behaviors by understanding the functional roles of their components.

We view mechanistic interpretability[1] as equivalent to extracting **causal graphs** explaining how intermediate NN computations mediate model outputs. This framing based in causality enables a new perspective of the field: prior surveys have organized

1 The meaning of "mechanistic interpretability" is debated; see Saphra and Wiegreffe (2024). We define the term as any study that aims to understand, explain, or modify a neural network's behavior by studying the model's internal components, such as its representations or weights.

the field according to methodological differences, whereas we taxonomize work in the field according to the kinds of causal mediators—or types of nodes in the causal graphs—that a study employs (e.g., neurons, non-basis-aligned directions, attention heads, and the like). We start by describing causal mediation analysis and its role in MI (§2); we also define the goals of MI, and goal-specific criteria by which the success of a mediator can be measured (§2.1). We then contrast this survey with other MI surveys (§3), noting in particular the lack of surveys that center the mediator type. Following this, we present a history of mechanistic interpretability for neural networks more broadly (§4), from backpropagation to the beginning of the current wave of mechanistic interpretability research.

We survey common mediators (units of causal analysis) used in mechanistic interpretability studies (§5), discussing the pros and cons of each mediator type. Should one analyze individual neurons? Combinations of neurons? Full activation vectors? More broadly, *what is the right unit of abstraction for analyzing and discussing neural network behaviors?* Any model component has pros and cons related to its level of granularity, whether it is a causal bottleneck, and whether it is natively part of the model (as opposed to whether it is learned via a separate module). The mediator type determines the kinds of methods that may be used to search over them; these search methods have their own pros and cons, which we use to organize the field in §6.

Finally, after surveying the field, we discuss practical considerations and implications for future work (§7). We point out mediators which have been underexplored, but have significant potential to yield new insights; propose future mediators that are likely satisfy the criteria laid out in §2.1; and suggest ways to measure progress in mechanistic interpretability moving forward. Figure 1 summarizes the content of this survey.[2]

2. Preliminaries

The counterfactual theory of causality. Lewis (1973) poses that a **causal dependence** holds iff the following condition holds:

> "An event E *causally depends* on C [iff] (i) if C had occurred, then E would have occurred, and (ii) if C had not occurred, then E would not have occurred."

Lewis (1986) extends the definition of causal dependence to be whether there is a **causal chain** linking C to E; a causal chain is a connected series of causes and effects that proceeds from an initial event to a final one, with potentially many intermediate events between them. This idea was later extended from a binary notion of whether the effect happens at all to a more nuanced notion of causes having influence on *how* or *when* events occur (Lewis 2000). Other work defines notions of cause and effect as measurable quantities (Pearl 2000); this includes direct and **indirect effects** (Robins and Greenland 1992; Pearl 2001), which are common metrics in causal interpretability studies.

Causal abstractions in mechanistic interpretability. **Causal graphs** are fundamental abstractions in the causality literature (Pearl 2000). A causal graph $\mathcal{H}$ is a directed acyclic graph consisting of nodes V and directed edges E. A **node** $V_i \in V$ corresponds to an action

2 Note that the methods discussed in this survey focus primarily on language models, so we will often use "language model" in place of "neural network" throughout this survey. However, most of the methods we discuss generalize beyond language models: most can, in theory, apply to any neural network.